Samyak Jain

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Education

University of Illinois Urbana-Champaign — Expected May 2028
Bachelor of Science, Physics and Mathematics, 4.0/4.0 GPA — *Champaign, IL*

- *Minors:* Computer Science, Quantum Information Science
- *Coursework:* Special Relativity, Graduate Dynamical Systems, Abstract Linear Algebra Honors, ODEs, PDEs, Fundamental Mathematics Honors, Intro Computer Science I & II, Data Structures
- *Activities:* Physics Research Seminar, Society of Physics Students, Physics Student Advisory Board

Technical Skills

Programming Languages: Python, C++, Java, Typst
Libraries & Software: Qiskit, NumPy, SciPy, Matplotlib, Pandas, gdspy, OpenGL, Ansys HFSS
Fabrication: Maskless photolithography, sputtering, AFM, SEM, wafer dicing

Experience

Quantum Hardware Researcher — Oct 2025 – Present
KouBit Lab — *Champaign, IL*

- Simulate coplanar waveguide structures using Ansys HFSS, achieving up to −30 dB suppression of slotline resonance at 15 GHz with integrated airbridges
- Design and fabricate airbridges to suppress parasitic slotline modes in coplanar waveguides using maskless photolithography
- Produce and characterize 30 μm × 70 μm × 3 μm airbridges via thin-film deposition, magnetron sputtering, wafer dicing, AFM, and SEM imaging

Quantum Information Theory Researcher — Sep 2024 – Mar 2025
Independent — *Remote*

- Built a modular quantum circuit simulation framework in Python (Qiskit) to model entanglement dynamics in noisy random Clifford circuits, validating and broadening theoretical predictions
- Designed and implemented library components for GHZ state initialization, circuit generation, depolarizing noise simulation, and logarithmic negativity computation
- Performed 20 parameter sweeps across 21 noise levels, applying R^2 and RMSE tests to validate results

Particle Physics Researcher — Jun 2024 – Sep 2024
DESY - Deutsches Elektronen-Synchrotron — *Hamburg, Germany*

- Conducted experiment on Smith-Purcell radiation from 6 GeV electron beam for beam diagnostics at DESY
- Reconstructed electron trajectories to enable electron tomography by utilizing 6 beam telescopes
- Presented research findings at the 13th Beam Telescopes and Test Beams Workshop

Projects

Chaotic Attractor Simulator | *TypeScript, WebGL, Vite, KaTeX*

- Built an interactive WebGL visualization for deterministic chaotic dynamical systems, simulating $100,000+$ particles evolving under RK4 integration at 60 fps
- Designed a cross-section slicer that filters particles in a $w \pm \varepsilon$ window, revealing 3D structure in a 4D system
- Visualized sensitive dependence on ICs by tracking perturbations and divergence in the Lorenz system

Hydrogen Atom Orbital Simulator | *C++, OpenGL, GLFW, Dear ImGui*

- Built a 3D hydrogen orbital visualizer using Monte Carlo rejection sampling of quantum probability densities across n, l, m states, rendering $100,000+$ particles in real time
- Designed ImGui-based control panel, orbit camera, zoom controls, and custom shaders

Nova Pset | *Typst*

- Developed a customizable problem set template using Typst, supporting reusable components for proofwriting, solutions, notes, and layouts
- Published package to Typst Universe to open-source utilities for other students

Distinctions

Beamline for Schools Winning Team **(2024)** · Massachusetts Science and Engineering Fair 3rd Place **(2025)** · James Scholar **(2026)** · National Merit Finalist **(2025)** · Massachusetts Boys State Delegate **(2024)**